

Scalability & Future Plan

Date	3 JULY 2026
Team ID	SWTID- 2026-9030
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users through the current Flask web application.	Deploy the application on IBM Cloud with load balancing and auto-scaling to support multiple concurrent users.

2	Data Storage	Uses the processed dataset and locally stored model files (model.pkl, scaler.pkl, encoder.pkl).	Integrate a cloud database such as IBM Cloud Databases, MySQL , or MongoDB for secure and scalable data storage.
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Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Prediction accuracy depends on the quality and completeness of the training dataset.	Incomplete or outdated data may lead to incorrect approval or rejection predictions.	High
2	The system predicts credit card approval only and does not consider real-time banking data or external credit bureau information.	Prediction results may not fully reflect an applicant's latest financial status.	Medium
3	The current application does not include user authentication or role-based access for administrators and analysts.	Limited security and restricted access control for sensitive applicant information.	Medium

Scalability Plan:

3	Performance	Prediction is performed using a locally loaded machine learning model..	Optimize model inference, enable API optimization, caching, and leverage IBM Watson Machine Learning for faster real-time predictions.
4	Security	Includes basic input validation in the Flask application.	Implement user authentication, role-based access control, HTTPS encryption, and secure API access to protect applicant data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on IBM Cloud	1 Month	Online access for users
Phase 2	Add user login and authentication	2 Months	Improved security
Phase 3	Improve model accuracy with more	3 Months	Better prediction accuracy
Phase 4	Add a multilingual interface	4 Months	Better user experience

